UNIT 2 · TOPIC 3(B)

Evaluation of Farm Machines

Syllabus topic: Farm machine system characteristics and evaluation

What to evaluate

Aspect	Measures
Functional performance	Quality of work (e.g., % weed kill, seed placement accuracy, threshing efficiency, % broken grain) and rate of work (capacity, ha/h or t/h)
Power/energy performance	Draft, torque, PTO power, specific energy (kWh/ha or kWh/t)
Durability & reliability	Accelerated endurance runs, breakdown records, MTBF
Safety and ergonomics	Unit 5
Economics	Cost/ha — Unit 3

Levels of testing

- Laboratory (component/test-rig, soil bin).
- Laboratory-field.
- Field tests per BIS test codes / **RNAM** test codes (Regional Network for Agricultural Machinery — standard procedures for seed drills, sprayers, threshers etc.).
- FMTTIs conduct confidential and commercial tests (procedures in Unit 3).

Practice MCQs

Q44. Quality of work of a seed drill is judged by:

(a) Fuel use
(b) Seed placement uniformity/depth accuracy
(c) Weight
(d) Draft only

Q45. RNAM test codes relate to:

(a) Tractor engines
(b) Standard test procedures for agricultural machines in Asia-Pacific region
(c) Chemical residues
(d) Irrigation design

Q75. MTBF stands for:

(a) Maximum time before failure
(b) Mean time between failures
(c) Minimum test bench force
(d) Mean torque at base frequency

Answers

Q44 — (b) Seed placement uniformity/depth accuracy
Q45 — (b) Standard test procedures for agricultural machines in Asia-Pacific region
Q75 — (b) Mean time between failures

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 1 (Research & Development), Ch. 2 (Field Capacities), §14.11 (Reciprocating Unbalance in a Mower); theory-of-machines standards (balancing, flywheels); tractor–implement mechanics

references.
